

Mountain Vista Water Co-op

Emergency Response Plan

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Cardston County, Alberta

This Emergency Response Plan (ERP) provides guidance to Mountain Vista Water Co-op leadership when faced with an emergency or unexpected situation. It defines roles, responsibilities, communication protocols, and emergency operating procedures for ten categories of incidents.

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1 General Information

1.1 Purpose of the Emergency Response Plan

The purpose of an Emergency Response Plan (ERP) is to provide guidance to MOUNTAIN VISTA Water Co-op (the Co-op) leadership when faced with an emergency or unexpected situation. The ERP should serve several useful purposes including:

- Promote reflection about what emergency situations could arise and planning to avoid or deal with these situations in advance.
- Identify an organization or chain of command that ensures the appropriate parts of the organization are informed and that decisions are taken with proper authority.
- Formalize a communication protocol within the organization and with key stakeholders such as regulators, customers, insurance company, suppliers, and contractors.
- Provide reference documents and information in a convenient format to aid in the response to an emergency.

1.2 What is an Emergency?

For the purposes of this ERP, an emergency is any incident or situation that has, or could result in:

- Personal injury to Co-op staff, customers or third parties resulting from Co-op operations;
- Interruption in the water supply to Co-op customers;
- Failure to meet water quality standards and consequent health risk to customers;
- Significant financial or reputational losses to the Co-op;
- Environmental damage; or
- Failure to comply with regulatory requirements.

Specific incidents that should be considered an emergency and which should result in the implementation of this ERP include:

- Accident resulting in personal injury as a result of, or related to, Co-op operations;
- Vehicle accident involving contractor or Co-op staff while engaged in Co-op operations;
- Failure of a bacteriological test performed by Alberta Health;
- Water quality analysis indicates an exceedance of chemical health standards for drinking water;
- Extended (more than 4 hours) loss of pressure in distribution system due to power failure or equipment breakdown;
- Fire, flood or other significant event at Co-op facilities that results in financial loss and potential extended loss of water supply to customers;

- Contamination of treated water in storage or in the distribution system that could pose a health hazard to customers;
- Equipment failure that could result in an extended loss, or reduction, of service to customers;
- Significant leak or rupture of distribution piping that requires immediate repair.

2 Organization, Roles, and Responsibilities

2.1 Emergency Response Team Organization

Role	Person
Co-op Chairman	Keith Wagner
Head Operator / First Responder	Shane McMorrow
Others as Required	Board Directors / Contractors

2.2 First Responder Roles and Responsibilities

The First Responder is any Co-op member or contractor who discovers the emergency. The First Responder should take the following actions:

- Don't Panic. Assess the situation, determine the problem, extent and action required to remedy.
- Do not enter a confined space or an otherwise risky area if you have safety concerns.
- Evacuate and call for help (raise the alarm) and notify a Co-op Director or the Head Operator.
- Call emergency services, as required.
- If applicable, take action to remove any injured person to a safe location, and administer first aid.
- Considering the nature of the emergency, take corrective action to limit further damage to life, environment, and property and to preserve evidence.
- Stay on site until contacted by the Head Operator or the Chairman of the Co-op with further instructions.

2.3 Chairman Roles and Responsibilities

The Chairman is responsible for all aspects of the response, including developing incident objectives and managing all incident operations. If the Chairman is not available, another Director from the Co-op Board will assume Chairman responsibilities.

- Establish immediate priorities especially the safety of responders, other emergency workers, bystanders, customers, and other people involved in the incident.
- Stabilize the incident by ensuring safety and managing resources efficiently and effectively.
- Ensure adequate health and safety measures are in place.
- In consultation with Head Operator, determine incident objectives and strategy. Approve the implementation of the written or oral Incident Action Plan.

- Establish and monitor emergency organization including assigning responsibilities to additional persons as necessary.
- Approve communications to Regulators and Customers in consultation with Head Operator.
- Work with Head Operator to compile an Incident Report and approve the report for distribution.

2.4 Head Operator Roles and Responsibilities

- On notification of a possible emergency, assess the on-site situation and confirm significance and type of emergency in consultation with the First Responder.
- Notify the Co-op Chairman of the emergency and any action taken and discuss remedial measures.
- Call and mobilize other operations staff and engage Contract Services as needed to assist in resolving the incident.
- Act as primary contact for Regulators including making initial contact with AEP and/or AHS to notify them of the emergency.
- Assist the First Responder, other operations staff, and Contract Services as necessary to assist in repairing equipment and returning water service to the community.
- In consultation with the Chairman, utilize the Customer Contact Procedure to advise customers of important information such as boil water advisories, limited use advisories or to inform them of the emergency and prognosis.
- Resume normal operations once incident is resolved.
- Work with the Chairman to compile a report of the emergency and file the final and approved report in the Co-op permanent records and with the Regulator as necessary to close the incident.

2.5 Person Responsible for Maintaining the ERP

The Board shall review the ERP annually or more frequently if there are changes in the Co-op operations and shall submit a revised ERP to the Chairman for approval. Once approved, The Board shall ensure distribution of the revised ERP to the Co-op Directors and to the Co-op facilities.

The Head Operator shall also ensure that an Operations Manual is kept up-to-date and available for use during an emergency. The Operations Manual must include at a minimum:

- Plans and Maps showing location of key equipment and valves.
- Equipment manuals for all key system equipment.
- Procedures for start up and shut down of key equipment and process units.
- Operations manual will be found in the pump shack containing maps, procedures, and contacts.
- A quick start guide will be posted on the wall of the pump shack.

3 Communications, Records and Reporting

3.1 Keeping Notes and Records During an Emergency

Documentation of the emergency, initial response and recovery in your log book is important; log sheets to be found in the pump shack. Each member of the emergency response team should keep a personal diary of their activities including:

- **When** — Exact time of the action and duration if applicable
- **Who** — including name, organization, position, cabin number
- **Where** — where is the person, how can they be contacted for follow-up
- **What** — details of the discussion, decisions taken, actions taken, results

3.2 Customer Notification Plan

On the decision of the Chairman (or his alternate), a message will be communicated to customers. When the decision is taken to issue a customer notification, the Head Operator will issue the approved communication as follows:

- **PRIMARY COMMUNICATION** — by email using the Customer contact list on file with the Co-op
- **SECONDARY COMMUNICATION** — Door to door or by phone call; Signs at key locations; WhatsApp Group, Email, Facebook Group

The Head Operator shall test the Primary Communication Protocol and update the Customer Contact List at least once per year.

3.3 Communication with Regulators

The Head Operator shall be the primary contact with AEP and AHS as necessary and in keeping with any regulatory requirements. Initial communication with regulators shall be made as soon as the site is physically secured and any immediate hazard is controlled. Any "Boil Water" notice should be discussed with AEP and AHS before issuance. The contact information for regulatory agencies is in Appendix A.

3.4 Incident Report Preparation and Filing

Within 48 hours of the end of the Emergency, the Chairman shall call a meeting to prepare a consolidated Incident Report. The Incident Report shall include:

- Chronology of events and activities
- Analysis of Causes
- Results and Outcomes

- Continuing activities
- Actions to prevent future incidents
- Any other information specifically required by a Regulator

4 Emergency & Contingency Plans

This section identifies each emergency and assesses possible damage and/or loss to the Co-op and its customers. Also included are emergency operating procedures to give guidance and help restore the water system.

4.1 Personal Injury or Vehicle Accident

Significance

- Financial loss, personal injury or environmental damage are direct consequences for the Co-op.
- Third Party Liability claims may result from an accident.

Action Needed

- Use 911 to call police, fire department or ambulance as required.
- First Responder or Head Operator will contact Chairman to initiate ERP.
- If accident is on Co-op property, immediately control risk — reduce pressure, shut down equipment, turn off power and other actions as appropriate.
- Control site access to prevent loss of evidence and facilitate access for emergency services.
- As soon as possible following accident documentation and investigation, take steps to resume normal, safe operations.
- Chairman shall contact AFRWC (insurance agent) to advise of an insurable incident and fill out Insurance Claim Forms as applicable.

4.2 Contamination Event

Significance

- A known or suspected contamination of source water or of water within the storage and distribution system could result in serious health issues for water users and liability for the Co-op.

Action Needed

- Head Operator or First Responder to take immediate action to remove source of contamination and if necessary to stop delivery of water.
- Head Operator or First Responder will contact The Board to initiate ERP.
- Head Operator shall advise AEP and AHS of the suspected or actual contamination event.
- If a boil water advisory or a do not use advisory is required, the Head Operator shall consult with Chairman and use the Customer Notification Plan in Section 3.2.

- Head Operator will consult with The Board and if appropriate, arrange for alternate safe water supply for customers.
- Once the source of the contamination is eliminated, Head Operator shall purge the system and test for adequate chlorine and absence of E. Coli.
- After consultation with The Board, the Head Operator shall use the Customer Notification Plan to rescind any boil water or do-not-use notifications.

4.3 Failure of Bacteriological Tests

Significance

- AHS should immediately inform the Co-op if any coliforms are present in a routine sample, indicating a possible health issue and need for remedial action.
- A failure could indicate an inappropriate test procedure or a true breakdown in water treatment.
- This testing will be completed by the County Operator.

Action Needed

- Head Operator notify AEP of the failure and notify the Chairman to initiate the Emergency Response Plan.
- The site from which the sample was taken shall be flushed and the chlorine residual at that point shall be checked to ensure it is greater than 0.10 mg/L.
- A second sample from the same site shall be taken to verify the failed test result. Samples shall also be taken from one site upstream and one site downstream.
- In addition, the Head Operator shall carry out the corrective actions as set out in the Alberta Environment Communication and Action Protocol For Failed Bacteriological Results In Drinking Water (Appendix C).
- If a boil water advisory is required, the Head Operator shall consult with Chairman and use the Customer Notification Plan in Section 3.2.
- Head Operator shall lead efforts to identify and remediate the cause of the contamination.
- On successful retesting, the Head Operator shall consult with the Chairman and advise all customers that the boil water advisory is over.

4.4 Loss of Water Supply / Well Failure

Significance

- A loss of water supply or well failure could result in a relatively long-term interruption in water supply to customers resulting in loss of confidence, and potential liabilities.

Action Needed

- Head Operator shall notify AEP of the failure and notify the Chairman to initiate the Emergency Response Plan.
- Head Operator, in consultation with Chairman, shall arrange for alternate water supply by truck or from another suitable source.
- The Head Operator shall consult with Chairman and use the Customer Notification Plan to advise all customers.
- Once the water supply is restored, Head Operator shall purge the system with treated water and test at system extremities for adequate chlorine and absence of E. Coli.
- After consultation with the Chairman, the Head Operator shall rescind any notifications to customers.

4.5 Power Supply Failure / Outage

Significance

- A power failure may result in a loss of pumping capacity and pressure reduction in the distribution system.
- An extended power failure could leave customers without water.
- In cold weather, protection of equipment against freezing may be required.

Action Needed

- Less than 4 Hours: Head Operator shall notify the power company and try to determine restoration time. If no significant leaks, monitor and ensure power is restored within 4 hours. Gas power generator to be powered up.
- Greater than 4 Hours: Head Operator shall notify the Chairman to initiate the ERP. Arrange for back up electrical power or back up pumps. If expected to last more than one day, notify AEP through the EDGE system and advise customers to restrict water use. Once power is restored, take chlorine sample and bacteriological test to confirm no contamination.

4.6 Flood, Fire or Natural Disaster

Significance

- A fire or natural disaster could cause financial damage to the Co-op and interrupt normal supply of safe water.

Action Needed

- Use 911 to call police, fire department or ambulance as required.
- Head Operator or First Responder shall contact Chairman to initiate ERP.
- If accident is on Co-op property, immediately control risk — reduce pressure, shut down equipment, turn off power as appropriate.

- If accident results in a Contamination Event, Loss of Supply, or Equipment Failure, proceed in accordance with those emergency plans as well.
- As soon as possible following natural disaster, take steps to resume normal, safe operations.
- Chairman shall contact AFRWC to advise of an insurable incident and fill out Insurance Claim Forms as applicable.

4.7 Low Chlorine Residual in Distribution System

Significance

- Low Chlorine residual in the distribution system is an indication of an equipment failure, change in water quality or other operational failure.
- A low chlorine residual could give rise to unsafe drinking water.

Action Needed

- If the bacteriological sample fails or if the low chlorine situation cannot be resolved, Head Operator shall consult with the Chairman and issue a Boil Water Notice using the Customer Notification Plan.
- Once the chlorine residual has been restored and any failed bacteriological tests have been re-tested, the Head Operator shall consult with the Chairman and rescind any notifications.

4.8 Water Not Meeting Drinking Quality Guidelines

Significance

- Drinking water must meet the established drinking water quality guidelines (DWQG) to ensure the short term and long-term health of water users.
- Any exceedance needs to be addressed with AEP and AHS as a non-compliance.

Action Needed

- Head Operator shall notify AEP and AHS of the exceedance and notify the Chairman to initiate the Emergency Response Plan.
- Normally, an immediate retest of the failed parameter is required to confirm that there is an exceedance.
- The Head Operator and Chairman shall consult with AEP and AHS and take necessary measures which may include issuing a "Do Not Use" notice, modifying the treatment process, or arranging for temporary water supply.
- Head Operator shall carry out corrective actions as set out in the Alberta Environment Action Protocol for Exceedances of Chemical Health Parameters (Appendix C).

4.9 Pipeline Leak or Break

Significance

- A pipeline leak or break can result in loss of water, interruption of service to customers and significant financial cost.
- A pipeline break can also lead to contamination of the distribution system.

Action Needed

- Head Operator shall notify AEP of the leak or break and notify the Chairman to initiate the Emergency Response Plan.
- Head Operator to assess the extent and location of the leak. If minor, advise the Chairman and handle as scheduled maintenance.
- If urgent repairs are required, Head Operator shall consult with Chairman and issue a notice of service interruption to customers in the affected area.
- Head Operator shall reduce pressure on the affected section but some pressure should be maintained to prevent contamination.
- Head Operator shall arrange for an excavator and repair crew, and locate adjacent pipelines, power and communication utilities before commencing excavation.
- Once repaired, Head Operator shall purge and disinfect the affected pipeline segment and arrange for chlorine and bacteriological sampling.
- Head Operator shall rescind any notices sent to customers and place the pipeline back into service.

4.10 Equipment Failure

Significance

- An equipment failure can result in a contamination event, a loss of supply event or a regulatory non-compliance.
- An equipment failure often results in significant financial cost to the Co-op.

Action Needed

- Head Operator shall notify the Chairman of the equipment failure and together they will determine the severity and best means of remediation.
- If the equipment failure will result in a loss of supply event, low chlorination, contamination or other emergency situation then Head Operator and Chairman shall follow those emergency procedures while arranging for repairs.
- If the equipment failure will result in a regulatory non-compliance but the Co-op has a temporary operating procedure that will reasonably protect water quality, then Head Operator shall report such non-compliance to AEP.

- Head Operator shall arrange for equipment repair or replacement as soon as practicable.
- Once repairs are completed, Head Operator shall notify Chairman to end the emergency.

A Appendix A — Contact List

Municipal Contacts	Phone	Email
Police / Fire / EMS / Spill Response	911	
Cardston County — Derek McCarthy (Water Operator)	403-332-0127	derek@cardstoncounty.com
Cardston County — Bruce Tishly	403-915-6899 403-653-4977	

AEP & AHS	Phone	Email
AEP Southern Region (Calgary)	403-297-7602	
AEP Emergency Line (EDGE)	1-800-222-6514	
AHS Environmental Public Health Officer	1-833-476-4743	

Insurance	Phone	Email
AFRWC — Lois Westacott	1-780-416-6540	lwestacott@fedgas.com
RMA — Debbie Depeel	1-780-955-4086	debora@rmainurance.com

Laboratories	Phone	Email
Maxxam Analytics Inc.	1-403-291-3077	
WHS Labs	1-403-250-9164	

Contractors & Utilities	Phone	Email
Excavator — Derrick Beck	403-308-5231	
ATCO GAS (Distribution)	1-800-511-3447	
Telus 24 Hour Emergency	1-888-811-2323	
Fortis 24 Hour Emergency	1-866-717-3113	
Alberta First Call	1-800-242-3447	

B **Appendix B — Copy of Water System Approval**

License to operate — on file with the Mountain Vista Water Co-op. Contact the Board for a copy.

C Appendix C — Protocols and Guidelines

C1 — How To Use Water During a Boil Water Advisory

See AHS site: How to use water safely during a boil water advisory

<https://albertahealthservices.ca/assets/news/advisories/ne-pha-2011-04-05-water-safety.pdf>

C2 — Action Protocol for Exceedances of Chemical Health Parameters

See AEP site: Action Protocol for Exceedances of Chemical Health Parameters in Drinking Water

<https://open.alberta.ca/dataset/78adf507-8b6e-405d-abb2-7bfd449b49fa/resource/4b52cea4-1fdd-4643-b285-e86bc7c0785a/download/2009-actionprotocolexceedancewater-2009.pdf>

C3 — Communication and Action Protocol for Failed Bacteriological Results

See AEP site: Communication and Action Protocol for Failed Bacteriological Results in Drinking Water

<https://open.alberta.ca/dataset/ed4019bd-9e4a-498b-8daf-62085e4d8520/resource/c6aba7bf-952e-4f7f-8ef3-84b6fa4c4e17/download/2009-commprotocolfailedbacterial-aug2009.pdf>

D Appendix D — Sample Public Notifications

MOUNTAIN VISTA Water Co-op WATER QUALITY ADVISORY

Effective date: _____

Used in situations in which the public health threat posed by the water supply system is modest, and actions can be taken to reduce the risks through means other than requiring a Boil Water Notice or Do Not Use Water Notice.

Until notified, users with weakened immune systems, or wishing additional protection should take precautions when drinking, washing fruits & vegetables, making beverages or ice, or when brushing teeth. Boil water rapidly for at least 1 minute, or use an alternate, safe source of water.

For more information contact: _____ at _____

MOUNTAIN VISTA Water Co-op BOIL WATER NOTICE

Effective date: _____

Used in situations in which the public health threat posed by the water supply system is significant and the nature of the threat is one that can be effectively addressed through boiling of the water.

Until notified all users are advised to: Bring water to a rolling boil for at least 1 minute, or use an alternate, safe source of water before using it for drinking, making ice, cooking, washing food, or brushing teeth. Use a hand sanitizer after washing hands.

For more information contact: _____ at _____

MOUNTAIN VISTA Water Co-op
DO NOT USE WATER NOTICE

Effective date: _____

Used in situations where a significant public health threat exists and the threat cannot be adequately addressed through a Water Quality Advisory or Boil Water Notice.

Until notified all users are advised to: Use bottled water or an alternate source of potable water for drinking, making ice, washing dishes, brushing teeth, and food preparation. **Boiling the water will not make it safe!** It is okay to use water for bathing and flushing toilets but please conserve water as much as possible.

For more information contact: _____ at _____